

- enoelektronski atom
- edini atom, za katerega lahko točno rešimo SSE
- dognanja pomembna tudi za večje atome

## OSN. PREDPOSTAVKE

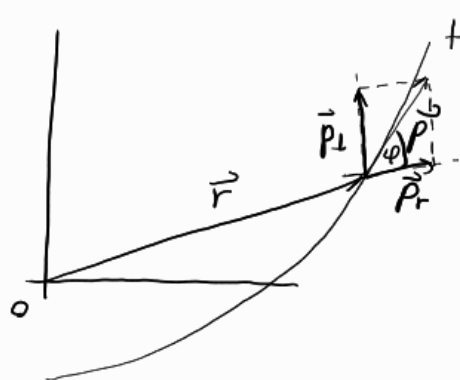
- en  $e^-$  v Coulombskem polju 1  $p^+$ , tadn formule izpeljujemo za naboj  $+ze_0$



$\rightarrow \begin{array}{c} x \\ + \end{array} \xrightarrow{r} e^- \quad \mu = \frac{M m_e}{M + m_e}$  reducirana masa

- problem vrtenja dveh teles prevedemo na vrtenje reducirane mase okoli razd. med delcema
- ker  $m_e \ll M$ :  $\mu \approx m_e$

potencial  $V(r) = -\frac{ze_0^2}{4\pi\epsilon_0 r} < 0$



trajektorija  $e^-$

$p_r = \mu \frac{dr}{dt}$

$p_{\perp} = \mu r \frac{d\varphi}{dt}$

$\vec{L} = \vec{r} \times \vec{p}$

$L = r p_{\perp} = r p \sin \varphi$

$\frac{\vec{p}^2}{2\mu} = \frac{\vec{p}_r^2 + \vec{p}_{\perp}^2}{2\mu} = \frac{\vec{p}_r^2}{2\mu} + \frac{\vec{L}^2}{2\mu r^2}$

namod. 2 operatorji:

$\hat{T} = \frac{\hat{p}^2}{2\mu} = \frac{1}{2\mu} \left( \hat{p}_r^2 + \frac{\hat{L}^2}{r^2} \right)$

rešiti moramo SSE v sferičnih koord.:

$$\underbrace{\left[ \frac{1}{2\mu} \left( \hat{p}_r^2 + \frac{\hat{L}^2}{r^2} \right) \right]}_{\hat{T}} \underbrace{\left[ -\frac{ze_0^2}{4\pi\epsilon_0 r} \right]}_{\hat{V}} \psi(r, \theta, \varphi) = E \psi(r, \theta, \varphi)$$

Nastavek:  $\Psi(r, \theta, \varphi) = R(r) Y(\theta, \varphi)$  • vstavimo v SSE, delimo z  $Y$ :

$$\frac{1}{2\mu} \vec{p}_r^2 R(r) - \frac{ze^2}{4\pi\epsilon_0 r} R(r) + \frac{1}{2\mu} \frac{R(r)}{Y(\theta, \varphi)} \hat{L}^2 Y(\theta, \varphi) = E R(r)$$

$$\vec{p}_r^2 \sim \nabla^2$$

$$\nabla^2 f(r) = \frac{d^2 f}{dr^2} + \frac{2}{r} \frac{df}{dr} = \frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{df}{dr} \right)$$

$$\hat{T} = \frac{\vec{p}_r^2}{2\mu} = -\frac{\hbar^2}{2\mu} \left( \frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) = -\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left( r^2 \frac{d}{dr} \right)$$

$$\text{lema: } \hat{L}^2 Y_{lm}(\theta, \varphi) = \hbar^2 l(l+1) Y_{lm}(\theta, \varphi)$$

$$\hat{L}_z Y_{lm}(\theta, \varphi) = m\hbar Y_{lm}(\theta, \varphi)$$

(rotator)

SSE:

$$-\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) - \frac{ze^2}{4\pi\epsilon_0 r} R(r) + \frac{1}{2\mu r^2} \hbar^2 l(l+1) R(r) = E R(r)$$

→ enačba za radialni del VF

1)  $l=0$  (od katerih neodv. reš. → sferično simetrična)

$$R(r) = A e^{-r/a} \rightarrow R'(r) = -\frac{A}{a} e^{-r/a} = -\frac{R}{a} \rightarrow R''(r) = \frac{R}{a^2}$$

$$-\frac{\hbar^2}{2\mu} \left( \frac{1}{a^2} - \frac{2}{r} \frac{1}{a} \right) R(r) - \frac{ze^2}{4\pi\epsilon_0 r} R(r) = E R(r)$$

če želimo, da velja  $\forall r$ , se morata in izničiti:

$$\frac{\hbar^2}{2\mu} \frac{1}{a} = \frac{ze^2}{4\pi\epsilon_0} \rightarrow a = \frac{4\pi\epsilon_0 \hbar^2}{ze^2 \mu} = \frac{m_e}{\mu} \frac{r_B}{z}$$

$$r_B = \frac{4\pi\epsilon_0 \hbar^2}{e^2 m_e} = 0,0529 \text{ nm}$$

BOHROV radij

a postavi skalo za ta problem  
(in za skoraj vso atomsko fiziko)

$$\text{• za H: } \frac{m_e}{\mu} \approx 1; z=1$$

$$\Rightarrow a = r_B$$

Energija:

$$E = -\frac{\mu}{m_e} z^2 E_0; E_0 = \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{m_e}{2\hbar^2} = 13,6 \text{ eV}$$

Rydbergova energija

## 2.) Neizotropne rešitve ( $l \neq 0$ ):

novi spremenljivki:  $r = a\rho = \frac{4\pi\epsilon_0\hbar^2}{Ze^2\mu} \rho$  ← brezdim. polmer

$$E = -\frac{\mu}{m_e} Z^2 \frac{E_0}{\hbar^2} \eta = -\frac{\hbar^2}{2ma^2} \eta \leftarrow \text{brezdim. energija}$$

$$\rightarrow \frac{d^2}{d\rho^2} (\rho R) + 2R - \frac{l(l+1)}{\rho} R = \eta \rho R$$

$$R(\rho) = \underbrace{\frac{F(\rho)}{\rho}}_{\text{polinom}} e^{-\sqrt{\eta} \rho}$$

mora biti regularno, ko  $\rho \rightarrow 0$  (če  $\rho \rightarrow 0 \Rightarrow F(\rho) \rightarrow 0$ )

$$F'' - 2\sqrt{\eta} F' + \left[ \frac{2}{\rho} - \frac{l(l+1)}{\rho^2} \right] F = 0$$

Robni pogoji:  $F(0) = 0$   
 $F(\infty) = 0$

$$F(\rho) = \text{polinom stopnje } n \text{ v } \rho \text{ (npr. } A\rho^n)$$

$$\Rightarrow \sqrt{\eta} = \frac{1}{n} \quad n \in \mathbb{N} \text{ če } l=0$$

$$\rightarrow E_{n00} = -\frac{\mu}{m_e} Z^2 \frac{E_0}{\hbar^2} \rightarrow \text{ustrezna VF: } \psi_{n00}(\vec{r}) = R_{n0}(r) Y_{00}(\theta, \varphi)$$

glavno kš ↓  
 $\underbrace{l \quad m}_{\text{izotropna reš.}}$

$$\psi_{n00}(\vec{r}, t) = \psi_{n00}(\vec{r}) e^{-i \frac{E_{n00} t}{\hbar}}$$

→ VF osn. stanja

$$R_{n0}(r) = \frac{F(r)}{r} e^{-\sqrt{\eta} r}$$

→ izkaže se, da so  $E$  odvisni samo od  $n$ , tudi če  $l \neq 0$

$$\boxed{E_{nl} = -\frac{\mu}{m_e} Z^2 \frac{E_0}{\hbar^2}} \quad n \in \mathbb{N}; \quad l = 0, 1, 2, \dots, \underline{n-1}$$

$$R_{nl}(\rho) = \frac{F_{nl}(\rho)}{\rho} e^{\rho/\eta} \quad ; \quad \rho = \frac{r}{a}$$

osn. stanje:  $n=1, l=0 \quad R_{10}(\rho) = \frac{2}{\sqrt{a^3}} e^{-\rho}$

1. vzb. st.:  $n=2, l=0 \quad R_{20}(\rho) = \frac{2}{\sqrt{2a^3}} \left(1 - \frac{\rho}{2}\right) e^{-\rho/2}$   
 $l=1 \quad R_{21}(\rho) = \frac{1}{2\sqrt{6a^3}} \rho e^{-\rho/2}$

2. vzb. st.:  $n=3, l=0 \quad R_{30}(\rho) = \frac{2}{3\sqrt{3a^3}} \left(1 - \frac{2\rho}{3} + \frac{2\rho^2}{27}\right) e^{-\rho/3}$   
 $\vdots$

normirni faktorji, da  
 $\int_0^\infty R_{nl}^2(r) \underbrace{r^2 dr}_{\text{od } dV} = 1$

$\hookrightarrow$  ničle polinoma, v bližini teh ničel zmanjšana verj. za detekcijo

Vse funkc. so regularne pri  $r=0 \Rightarrow$  heničelna verj., da e najdemo "v protonu!"

glavno kš  $n=1, 2, 3, \dots$

kš tirne VK  $l=0, 1, \dots, n-1$

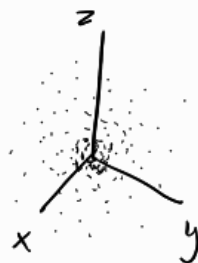
projekcija na z os  $|m| \leq l$

Degeneracija:  $\sum_{l=0}^{n-1} (2l+1) = n^2$

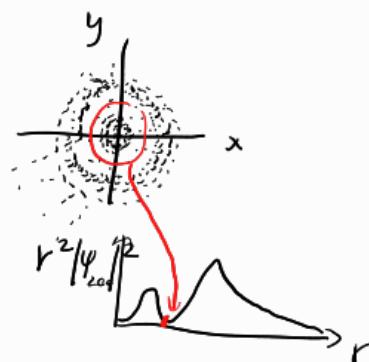
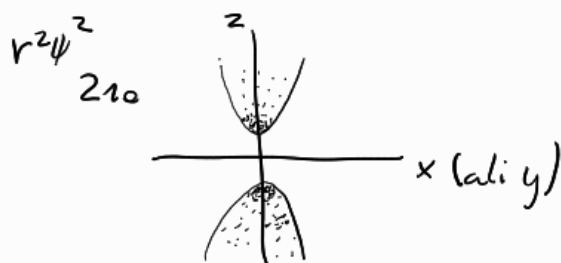
VF za najnižja kš:

$n=1; l=0; m=0$

$\psi_{100}(\vec{r}) = \underbrace{R_{10}(r)}_{n,l} \underbrace{Y_{00}(\theta, \phi)}_{l,m} \rightarrow |\psi_{100}|^2$

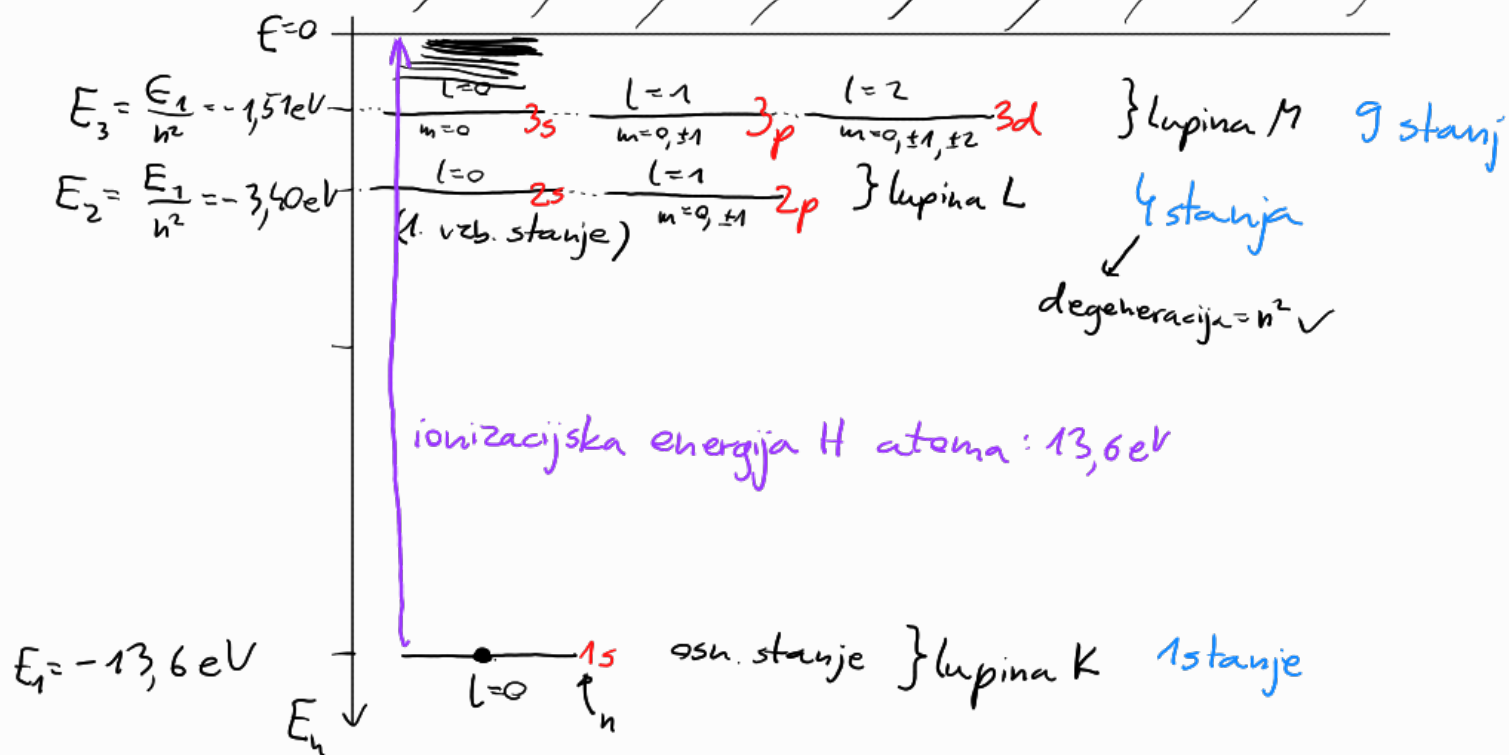


$\psi_{200}(\vec{r}) = R_{20}(r) Y_{00}(\theta, \phi) \rightarrow$



# Nivjska shema H

Kontinuum



| l | črka/oznaka     |
|---|-----------------|
| 0 | s (sharp)       |
| 1 | p (principal)   |
| 2 | d (diffuse)     |
| 3 | f (fundamental) |

$$\langle r \rangle = \int \Psi_{nlm}^* r \Psi_{nlm} dV = \int |\Psi_{nlm}|^2 r dV =$$

$$= \int R_{nl}^2 r^2 dr = a n^2 \left\{ 1 + \frac{1}{2} \left[ 1 - \frac{l(l+1)}{n^2} \right] \right\}$$

$\int |\Psi_{nl}|^2 d\Omega = 1$  za H atom  $r_B$

za  $n=1$ :  $\langle r \rangle = \frac{3}{2} r_B$

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{a n^2} \longrightarrow \langle V \rangle = -\frac{z e_0^2}{4\pi \epsilon_0} \left\langle \frac{1}{r} \right\rangle = -z^2 \left( \frac{e_0^2}{4\pi \epsilon_0} \right)^2 \frac{\mu}{\hbar^2 n^2}$$

$$\left\langle \frac{1}{r^2} \right\rangle = \frac{2}{a^2 n^3 (2l+1)} \quad \text{podobno } \langle E \rangle = \frac{\langle V \rangle}{2}$$

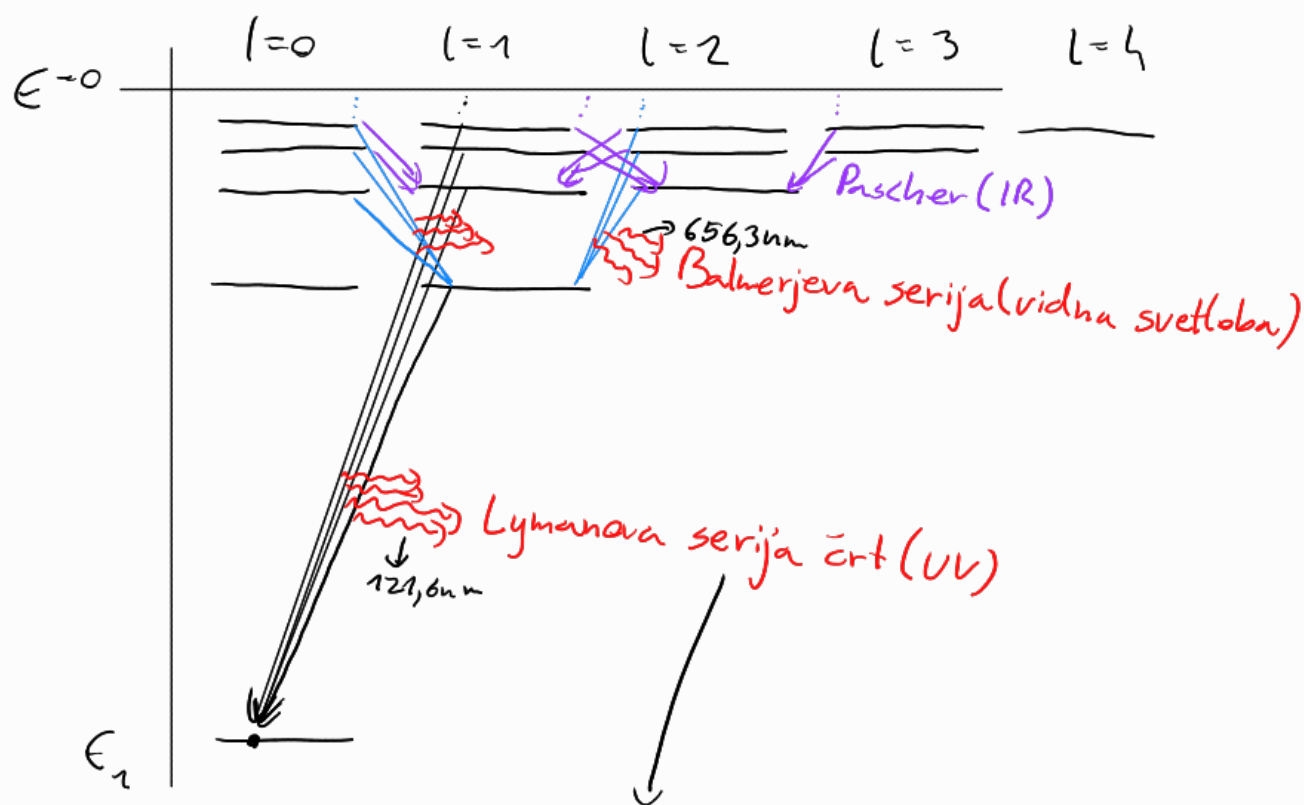
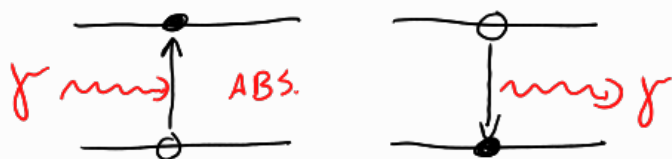
$$\left\langle \frac{1}{r^3} \right\rangle = \frac{2}{a^3 n^3 (l+1)(2l+1)}$$

Sevalni spekter: foton odnese razliko med  $E_n$  in  $E_{n'}$  ob prehodu  $n \rightarrow n'$

(obraten proces (absorpcija): foton prinese točno  $E_n - E_{n'}$ )

$$h\nu = \frac{hc}{\lambda} = E_n - E_{n'} = -E_0 \left( \frac{1}{n^2} - \frac{1}{n'^2} \right) > 0 \rightarrow n' > n$$

$= E_n$



lahko izmerimo zelo natančno:

Pri prehodih se mora  $l$  spremeniti za  $\pm 1$

$$\Delta l = \pm 1$$

12 BIRNO PRAVILO

$$\frac{1}{\lambda} = - \frac{E_0}{hc} \left( \frac{1}{n^2} - \frac{1}{n'^2} \right)$$

$R_y = 10977576 \text{ m}^{-1}$  (Rydbergova konst.)

(če vzamemo  $m_e = \mu$ :  $R_y = 10973731,5 \text{ m}^{-1}$ )

# Atom v (zunanjem) magnetnem polju

(→ magnetne interakcije in spin)

TIRNA VK → tirni magn. moment

Klasično:

dolž. gostota naboja po  $\odot \cdot e = -\frac{e_0}{2\pi r}$

$$\text{tok } I = -\frac{e_0 v}{2\pi r}$$

$$\text{mag. moment} = \mu = -\frac{e_0 v}{2\pi r} \cdot \pi r^2 = -\frac{e m_e v r}{2 m_e}$$

Kvantno:

$$\hat{\mu} = -\frac{e_0}{2m_e} \hat{L} = -\frac{e_0 \hbar}{2m_e} \frac{\hat{L}}{\hbar} = \boxed{\text{OPERATOR MAG. MOMENTA}} \quad \begin{matrix} -g_L \mu_B \frac{\hat{L}}{\hbar} ; \mu_B = \frac{e_0 \hbar}{2m_e}, g_L = 1 \end{matrix}$$

$$\boxed{\hat{\mu}_z = -g_L \mu_B \frac{\hat{L}_z}{\hbar}}$$

Bohrov magneton

$$\mu_B = 9,27 \cdot 10^{-24} \text{ J/T}$$

giromagn.  
faktor za  
tirno VK

$$\mu = \langle |\hat{\mu}| \rangle = g_L \mu_B \frac{\sqrt{\hbar^2 l(l+1)}}{\hbar} = g_L \sqrt{l(l+1)} \mu_B$$

$$\mu_z = \langle \hat{\mu}_z \rangle = g_L \mu_B \frac{m_l \hbar}{\hbar} = g_L m_l \mu_B$$

KLASIČNO:

$\vec{\mu}$  dano v zun. polje  $\vec{B}$ :

$$\frac{d\vec{\Gamma}}{dt} = \vec{\mu} \times \vec{B} = -\frac{e_0}{2m_e} \vec{\Gamma} \times \vec{B} = \vec{\omega} \times \vec{\Gamma}, \quad \vec{\omega} = \frac{e_0}{2m_e} \vec{B} = \text{konst.}$$

→ precesija  $\vec{\Gamma}$  okrog  $\vec{B}$  s frekvenco  $\boxed{\frac{\omega_L}{2\pi} = \frac{e_0 B}{4\pi m_e}}$

Larmorjeva prec. frekv.

Kvantno to ni ok opis.

$$E_{\text{mag}} = -\vec{\mu} \cdot \vec{B} = -\mu_z B \quad \text{če } B \parallel z\text{-os}$$

$$\hat{H}_{\text{mag}} = -\hat{\vec{\mu}} \cdot \vec{B} = -\hat{\mu}_z B$$

$$\rightarrow \boxed{\hat{H} = \hat{T} + \hat{V} + \hat{H}_{\text{mag}}}$$

$$\langle \hat{H}_{\text{mag}} \rangle = E_{\text{mag}} = g_l m_l \mu_B B$$

$$E_{nlm} = -\frac{E_0}{n^2} + g_l m_l \mu_B B$$

ZEEMANOV POJAV

nivoji se razcepijo



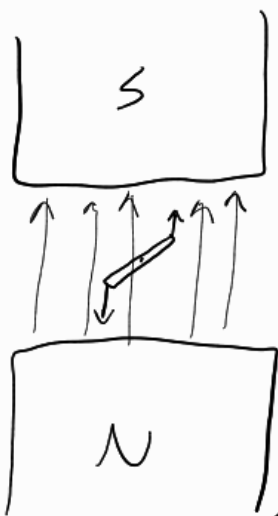
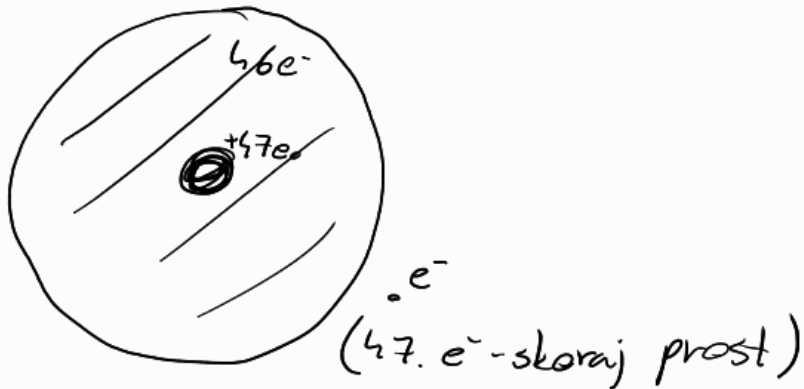
npr.  $l=1 \rightarrow m_l = 0, \pm 1$

degeneracija  $2l+1 = 2(n-1)+1 = 2n-1$   
LHO

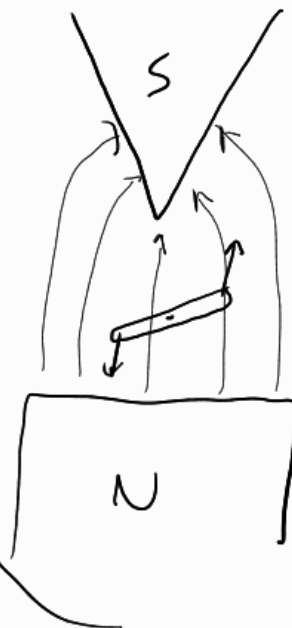
# Stern-Gerlachov poskus (spin $e^-$ )

- pokazala sta, da v  $B \neq 0$  osn. stanje razcepi na 2 nivoja
- očitno nek drug tip VK, ne tirna

atom Ag:



$\vec{B}$  homogeno  
 $\Rightarrow \sum \vec{F} \approx 0$   
 $\vec{M} \neq 0$



$\vec{B}$  nehomogeno  
 $\Rightarrow \sum \vec{F} \neq 0$   
 $\vec{M} \neq 0$   
izkoristila S in G



os = z :

$$\vec{F}_z = - \frac{\partial \epsilon_{mag}}{\partial z} = \mu_z \frac{\partial B}{\partial z}$$



kolimator  
(dobro definira smer)



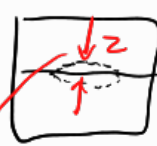
pečica

$T \approx 1000^\circ\text{C}$

Ag pare



$B=0$



$B \neq 0$

2 packi

Hitrost Ag atomov:  $\frac{3}{2} k_B T = \frac{1}{2} M v^2$

merimo odklon

$$\langle \vec{z} \rangle = \frac{1}{2} a t^2 = \frac{1}{2} \frac{F_z}{m} \left( \frac{y}{v} \right)^2 = \frac{1}{2} \frac{1}{M} \mu_z \frac{\partial B}{\partial z} \left( \frac{y}{v} \right)^2$$

?

$\rightarrow \boxed{\mu_z = \pm \mu_B}$  ugotovitev

očitno ne gre za tirno VK, saj bi morala videti 1, 3, 5, ... pad

$\rightarrow$  tej VK pravimo SPIN

Tirna VK

$$\langle \vec{L}^2 \rangle = \hbar^2 L(L+1) \quad \langle L_z \rangle = m_L \hbar \quad |m_L| \leq L$$

Spinska VK:

$$\langle \vec{S}^2 \rangle = \hbar^2 s(s+1) \quad \langle S_z \rangle = m_s \hbar \quad |m_s| \leq s$$

edina možnost: s polštevilski

$\vec{K} \vec{S}$  spina

$s = \frac{1}{2}$

$m_s = \pm \frac{1}{2}$

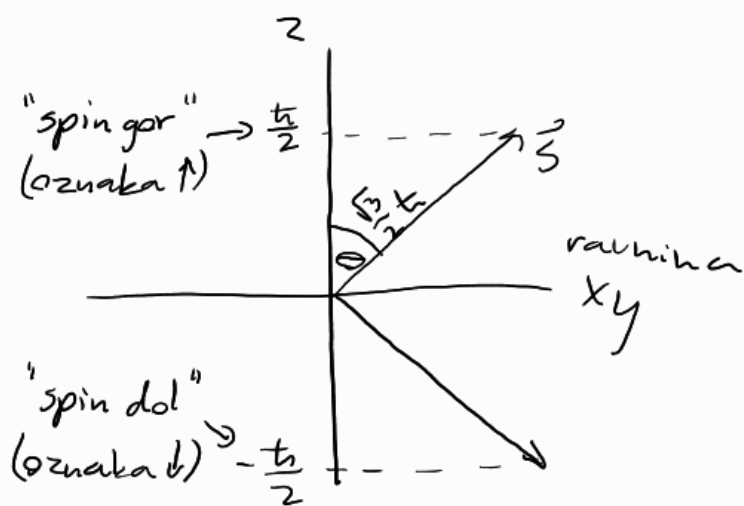
$\vec{K} \vec{S}$  projekcije spina na z-os

2 vidika operatorjev:

$$\vec{S}^2 \chi_{sm_s} = \hbar^2 s(s+1) \chi_{sm_s} = \frac{3}{4} \hbar^2 \chi_{sm_s}$$

$$\vec{S}_z \chi_{sm_s} = m_s \hbar \chi_{sm_s} = \pm \frac{\hbar}{2} \chi_{sm_s}$$

ne deluje na NOBENE KOORDINATE



$$\theta = \arccos \frac{1/2}{\sqrt{3}/2} = 54,7^\circ$$

"gor" in "dol" se lahko nanaša tudi na  $\vec{\mu}_s \propto -\vec{S}$

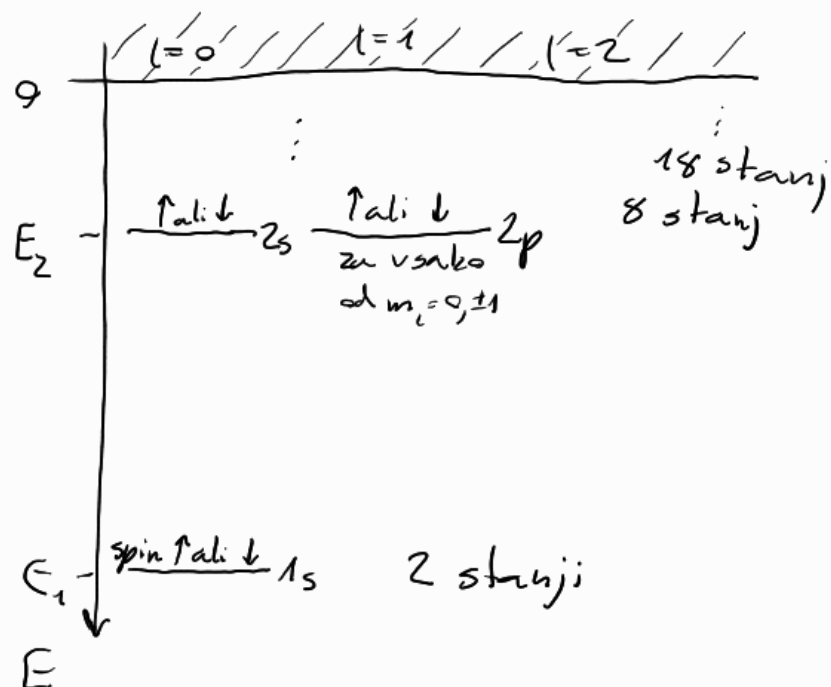
$\rightarrow$  VF  $e^-$  v H-atomu

$$\psi_{nlm_l s m_s} = R_{nl}(r) Y_{lm_l}(\theta, \varphi) \chi_{sm_s} e^{-iE_n t / \hbar}$$

lahko izpuščino, ker vedno  $s = \frac{1}{2}$

samo  $\uparrow$  ali  $\downarrow$

$\rightarrow$  nova nivojska shema!



Degeneracija  $2n^2$

Kaj se spremeni pri mag. momentih?

• tirni:  $\vec{\mu}_L = -g_L \mu_B \frac{\vec{L}}{\hbar}$   $g_L = 1$  (giromag. faktor za tirna VK)

• spinski:  $\vec{\mu}_S = -g_S \mu_B \frac{\vec{S}}{\hbar}$   $g_S = 2$  (giromag. fakt. za spinsko VK)

$\langle \mu_{S,z} \rangle = g_S \mu_B \frac{m_s \hbar}{\hbar} = m_s g_S \mu_B$

v poskusu sta SSG videla, da  $\langle \mu_{S,z} \rangle \approx \pm \mu_B \Rightarrow m_s g_S = \pm 1 \Rightarrow \underline{g_S = 2}$

V resnici  $g_S$  ni točno 2:

$$\frac{\mu_e}{\mu_B} - 1 = \frac{g_S - 2}{2} = 0,00115965218091 \pm (0,26 \cdot 10^{-12})$$

Eden od triumfov fizike

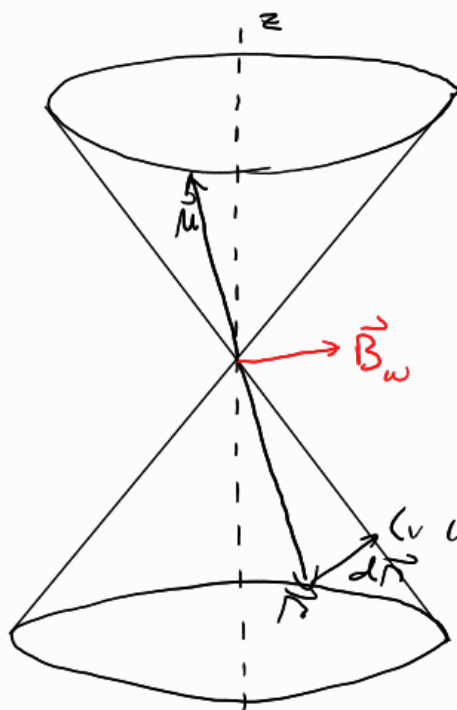
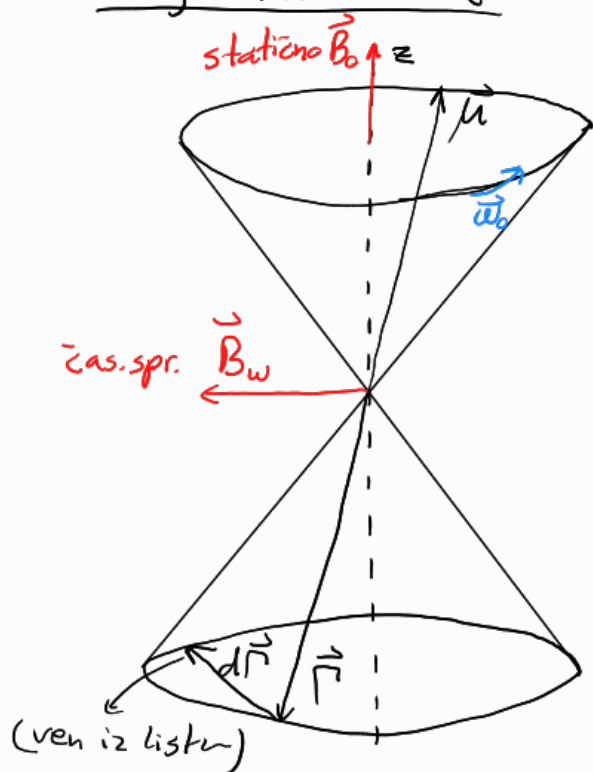
prej:  $\vec{\mu}_L \parallel \vec{L}$  in  $\vec{\mu}_S \parallel \vec{S}$

$\vec{J} = \vec{L} + \vec{S}$  celotna VK

$$\underbrace{\vec{\mu}_L + \vec{\mu}_S}_{\text{celotni magn. moment}} = -\frac{\mu_B}{\hbar} (\vec{L} + 2\vec{S}) \parallel \vec{L} + \vec{S}$$

celotni  
magn.  
moment

Magn. resonanca:



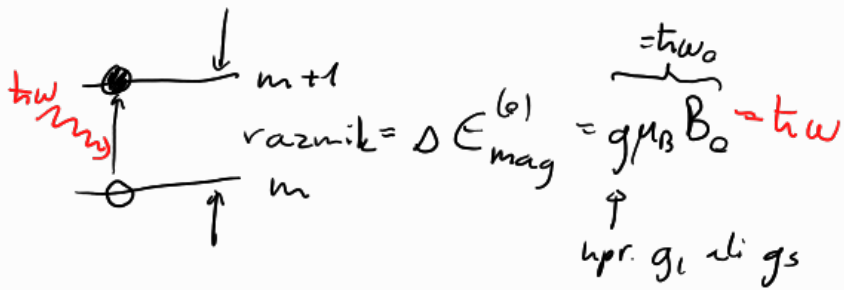
$$\omega_0 = \frac{e_0 B_0}{2m_e}$$

$$E_{\text{nag}}^{(s)} = -\vec{\mu} \cdot \vec{B}_0$$

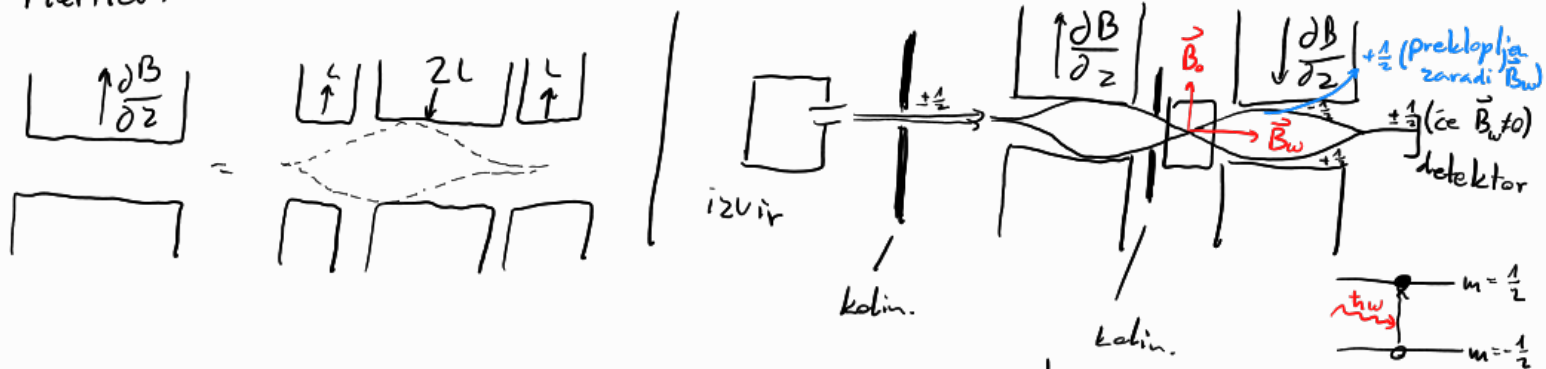
zaradi  $\vec{B}_w$  imamo  
dodatni navor  $\vec{\mu} \times \vec{B}_w$

Če v sinhronizirani s  
precesijo VK okoli  $\vec{B}_0$ ,  
lahko  $\vec{r}$  spravimo  
k drugim kotom  
→ zasukanol!

→ Doseči moramo prehod med 2 mag. podstanjema



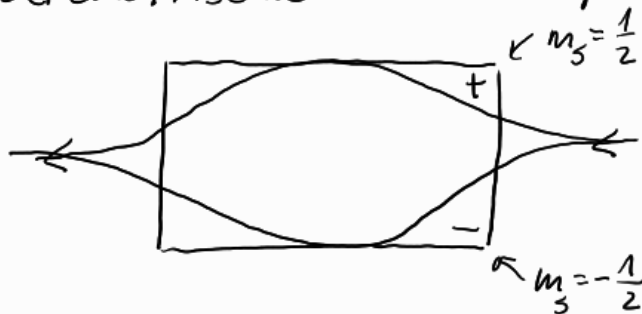
Meritev:



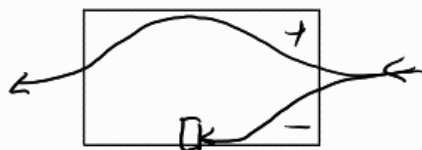
jakost pade, ker eno komp. spinske VK odstranimo  
• položaj minimuma lahko določimo zelo natančno

## Še nekaj (miselnih) eksperimentov s spinom $\frac{1}{2}$

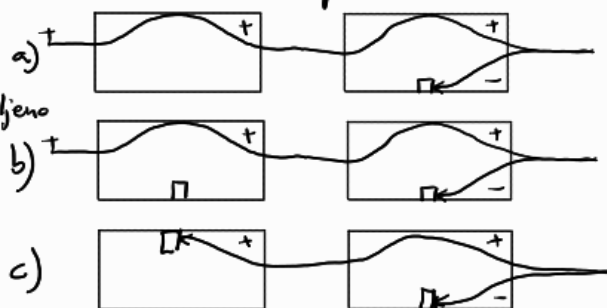
• beremo/rižemo od desne proti levi (običajno v KM)



Polarizator (filter):



analizator polarizator



• za študij prepustnosti def. c kolizino  $\chi(d, p)$

• verj. za detekcijo  $|\chi(d, p)|^2$

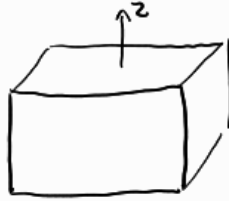
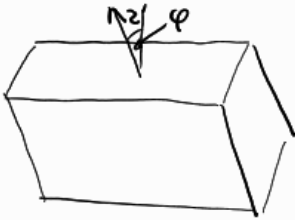
beremo

iz b), c) razberemo:  $|\chi(+z, +z)|^2 = 1$

$$|\chi(-z, +z)|^2 = 0$$

$$|\chi(+z, -z)|^2 = 0 \quad |\chi(-z, -z)|^2 = 1$$

Analizator zavrtimo za  $\varphi$  glede na polarizator (okoli osi prenikanja delcev)



Pričakujemo:

$$|\chi(+z', +z)|^2 \neq 1$$

$$|\chi(-z', +z)|^2 \neq 0$$

$$\text{Toda: } |\chi(+z', +z)|^2 + |\chi(-z', +z)|^2 = 1$$

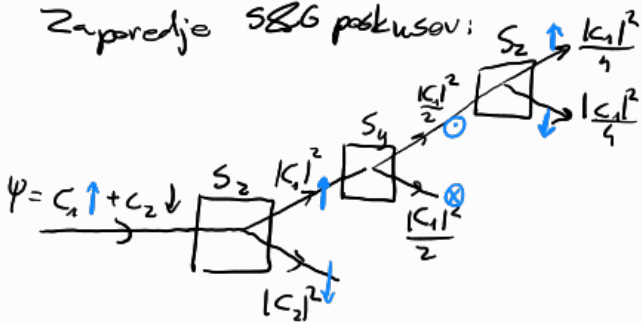
$$|\chi(-z', -z)|^2 + |\chi(+z', -z)|^2 = 1$$

Uganemo:

$$|\chi(+z', +z)|^2 = |\chi(-z', -z)|^2 = \cos^2 \frac{\varphi}{2}$$

$$|\chi(-z', +z)|^2 = |\chi(+z', -z)|^2 = \sin^2 \frac{\varphi}{2}$$

Zaporedje SSG poskusov:



Vsak žarek v trenutnem SSG "pozabi", kaj se mu je zgodilo v prejšnjem.

### Seštevaje VK

$$\boxed{\vec{J} = \vec{L} + \vec{S}} \text{ celotna VK}$$

$\vec{L}$  in  $\vec{S}$  kvantizirana  $\Rightarrow$  tudi  $\vec{J}$  naj bi bila kvantizirana

Iz geometrije (trikotniška neen.):  $||\vec{L}| - |\vec{S}|| \leq |\vec{J}| \leq |\vec{L}| + |\vec{S}|$

$\vec{L}, L_z$ : lastne vrednosti so  $\hbar^2 l(l+1)$  oz.  $m_l \hbar$

$\vec{S}, S_z$ : lastne vrednosti so  $\hbar^2 s(s+1)$  oz.  $m_s \hbar$   
 $\frac{3}{4} \hbar^2$  za  $e^-$      $\frac{1}{4} \hbar^2$  za  $e^+$

$\rightarrow \vec{J}, J_z$ : lastne vrednosti  $\hbar^2 j(j+1)$  oz.  $m_j \hbar$ ;  $j$  KŠ celotne VK

$m_j$  KŠ projekcije celotne VK na os  $z$

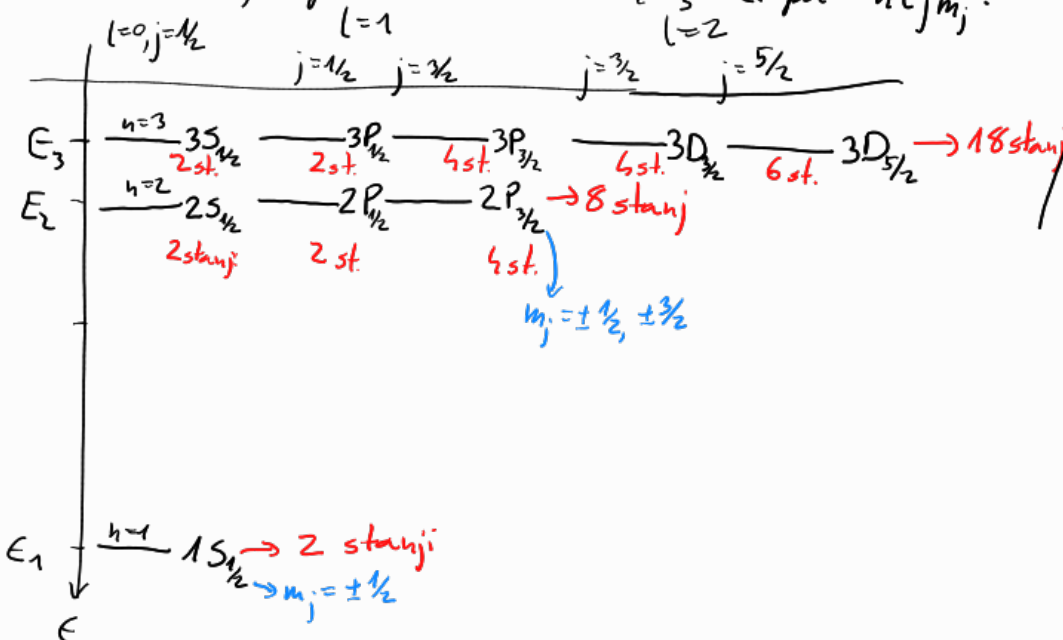
očitno  $m_j = m_l + m_s$  (ker  $J_z = L_z + S_z$ )

$l = 0, 1, \dots$ ;  $s = \frac{1}{2} \Rightarrow j$  polštevilski (najmanj  $\frac{1}{2}$ )

$m_j$  polštevilski

Najpreprostejša možnost:  $l=0, s=\frac{1}{2} \Rightarrow j=\frac{1}{2}$   
 $l \neq 0, s=\frac{1}{2} \Rightarrow j=(\pm \frac{1}{2})$

• ekvivalentno je povedati KŠ:  $n(l m_l m_s)$  ali pa  $n(l j m_j)$ :



Spektroskopske oznake:

$2s+1$   
 $n$   
 $j$

glavna KŠ | KŠ celotne VK

črke, ki pove l:

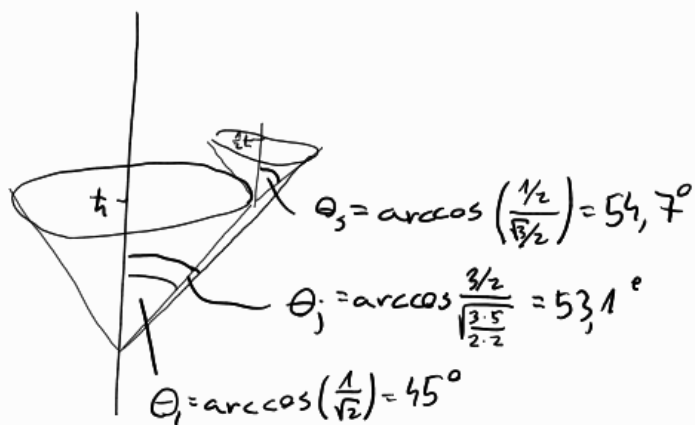
- S ...  $l=0$
- P ...  $l=1$
- D ...  $l=2$
- F ...  $l=3$
- ...

$$||\vec{L}| - |\vec{S}|| \leq |\vec{J}|$$

$$\vec{J}^2 \geq (\vec{L} - \vec{S})^2 = \vec{L}^2 + \vec{S}^2 - 2\vec{L} \cdot \vec{S}$$

$$\hbar^2 j(j+1) \geq (\sqrt{\hbar^2 l(l+1)} - \sqrt{\hbar^2 s(s+1)})^2 = \hbar^2 l(l+1) + \frac{3}{4}\hbar^2 - 2\sqrt{l(l+1)}\sqrt{\frac{3}{4}}$$

$\rightarrow$  z vstavljanjem  $j = (\pm \frac{N}{2})$   $N = \pm 1, \pm 3, \pm 5, \dots$  ugotovimo, da sta možni samo  $j = (\pm \frac{1}{2})$ .



## Seštevanje ali sklapljanje katerih koli v.k.

npr.  $\vec{L}_1, \vec{L}_2$

$$\vec{L} = \vec{L}_1 + \vec{L}_2 \rightarrow L_z = L_{1z} + L_{2z} \quad \checkmark$$

$$\vec{L}^2 = (\vec{L}_1 + \vec{L}_2)^2 = \vec{L}_1^2 + \vec{L}_2^2 + 2(\underbrace{L_{1x}L_{2x} + L_{1y}L_{2y} + L_{1z}L_{2z}}_{\text{niso ostro določene}})$$

$$L \longleftrightarrow L_1 \quad L_2$$

en par  $L_1, L_2$  ne določa

$L$ -ja enolično

$$\rightarrow \psi_{lm} = \sum_{l_1, l_2} C_{l_1 m_1 l_2 m_2}^{lm} \psi_{l_1 m_1 l_2 m_2}$$

$m = m_1 + m_2$

Clebsch-Gordanov koeficient

Zgled:

$$\psi_{10} = C_{10}^{\sqrt{\frac{3}{10}}} \psi_{1121} + C_{10}^{-\sqrt{\frac{2}{5}}} \psi_{1020} + C_{10}^{\sqrt{\frac{3}{10}}} \psi_{112-1}$$

$1-1=0 \checkmark$

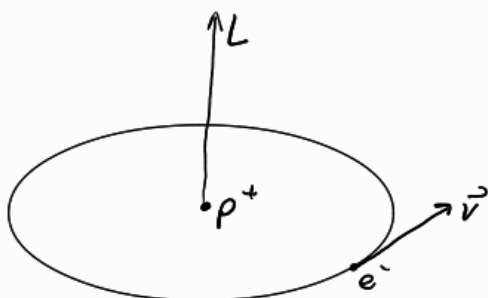
2 ↑ 1  
1 ↓ 1

• koeficienti so taki, da imamo normalizacijo na 1:

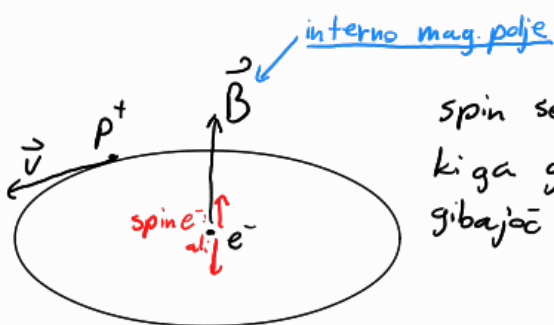
$$\frac{3}{10} + \frac{2}{5} + \frac{3}{10} = 1 \checkmark$$

## Sklopitev spin-tir (" $L$ ", " $\vec{L} \cdot \vec{S}$ ", ...)

- relativistični efekt
- za predstavo: Bohra model



v sistemu  $e^-$ :



spin se znajde v  $\vec{B}$ ,  
ki ga generira  
gibajoč naboj

$$E_{\text{mag}} = -\vec{\mu} \cdot \vec{B}$$

(za  $e^-$  je  $\vec{\mu} \parallel -\vec{S}$ )

→ stanja z enakim  $j$  in razl.  $j = l \pm 1/2$  bodo imela  
malenkostno drugačno  $E$ .

The diagram illustrates the Zeeman effect in a 3p state. On the left, an energy level diagram shows the splitting of the 3p state ( $l=1$ ) into three levels:  $2P_{3/2}$ ,  $2P_{1/2}$ , and  $2P_{-1/2}$ . The energy difference between the original 3p state and the 1s state ( $l=0$ ) is  $13.6 \text{ eV}$ . The energy difference between the split levels is  $\Delta E \approx 10^{-4} \text{ eV}$ . The splitting is labeled "čvrsta linija" (strong line) and "moćno pretirano" (strongly exaggerated). On the right, two vector diagrams show the orientation of the magnetic moment  $\vec{\mu}$  and the magnetic field  $\vec{B}$ . The top diagram shows  $\vec{\mu}$  and  $\vec{B}$  aligned, corresponding to  $\vec{\mu} \cdot \vec{B} > 0$ . The bottom diagram shows  $\vec{\mu}$  and  $\vec{B}$  anti-aligned, corresponding to  $\vec{\mu} \cdot \vec{B} < 0$ .

- ker se  $\vec{p}$  giblje (in ima statično el. polje  $\vec{E}$ ), generira tudi  $\vec{B} = 0$  (LT)

$$\vec{B}_{int} = -\frac{\vec{v} \times \vec{E}}{c^2} = \frac{Ze_0}{4\pi\epsilon_0} \frac{\vec{r} \times \vec{v}}{c^2 r^3} \frac{me}{me}$$

od tu ime sklopita  
A Ls

• Spin. mag. moment  $e^-$  čuti  $\vec{B}_{int}$ :  $E_{ls} = -\vec{\mu}_s \cdot \vec{B}_{int} = \left( g_s \mu_B \frac{\vec{S}}{\hbar} \right) \cdot \left( \frac{Ze_0}{4\pi\epsilon_0} \frac{\vec{L}}{m_e c^2 r^3} \right) = \frac{Ze_0}{4\pi\epsilon_0} \frac{\vec{L} \cdot \vec{S}}{m_e c^2 r^3}$

$\vec{\mu}_s = -g_s \mu_B \frac{\vec{S}}{\hbar}$   
 $\parallel$   
2  $\parallel$   
 $\frac{e\hbar}{2m_e}$

→ mehanika "Thoma sava  $\frac{1}{2}$ " (relativistični račun)

→ majika "Thomasova  $\frac{1}{2}$ " (relativistični račun)

$$\Rightarrow \boxed{\epsilon_{L3} = \frac{Ze^2}{4\pi\epsilon_0} \frac{\vec{L} \cdot \vec{S}}{2m_e^2 c^2 r^3}}$$

Klasifikacija stanj:  $\psi_{n(m,s)}$   
                                 dobri KS  
 - boje ikarati lahko ostro določimo

ko upoštevamo sklopitev LS:  $\psi_{n(l)m_i}$

$\vec{L} \cdot \vec{S}$  fixera kot med  $\vec{L}$  in  $\vec{S}$ , då ne precessirar mer utdragna.



- obroč z-osi precesira  $\vec{J}$  (in ima ostre projekcije  $m_j$ )

- Kdaj upoštevamo sklepiter LS in kdaj ne?

Zaradi sklopitve LS se stanja z istima  $n, l$  todarazl. j razcepijo.

→ posledica je fina struktura spektralnih črt (oz. energ. nivojev)

• brezdim. količina, ki postavi ustrezno skalo:

$$\alpha = \frac{e_0^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137} \quad \text{Konstanta fine strukture}$$

• efekti so  $O(\alpha^2) \approx \frac{1}{10^5}$

$$\begin{aligned}\hat{S}_z &= m_s \hbar \\ \hat{L}_z &= m_l \hbar \\ \hat{J}_z &= m_j \hbar\end{aligned}$$

$$\rightarrow E_{ls} = Z\alpha \frac{\hbar}{2m_e^2 c} \frac{\vec{L} \cdot \vec{S}}{r^3}$$

$$\vec{J} = \vec{L} + \vec{S}$$

$$\vec{J}^2 = \vec{L}^2 + \vec{S}^2 + 2\vec{L} \cdot \vec{S}$$

$$\vec{L} \cdot \vec{S} = \frac{1}{2}(\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$$

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{2}{a^3 n^3 l(l+1)(2l+1)}$$

$$\langle E_{ls} \rangle = \int \psi_{nljm_j}^* - \frac{\vec{L} \cdot \vec{S}}{r^3} \psi_{nljm_j} dV =$$

$$= Z\alpha \frac{\hbar}{2m_e^2 c} \frac{\langle \vec{J}^2 \rangle - \langle \vec{L}^2 \rangle - \langle \vec{S}^2 \rangle}{2} \left\langle \frac{1}{r^3} \right\rangle = Z\alpha \frac{\hbar}{4m_e^2 c} \left[ \hbar^2 j(j+1) - \hbar^2 l(l+1) - \hbar^2 \frac{3}{4} \right] \left\langle \frac{1}{r^3} \right\rangle =$$

$$= Z\alpha \frac{\hbar^3}{4m_e^2 c} \left[ j(j+1) - l(l+1) - \frac{3}{4} \right] \left( \frac{Z\alpha m_e c}{\hbar} \right)^3 \frac{2}{l(l+1)(2l+1)} =$$

$$= \frac{Z^4 \alpha^4}{2\hbar^3} m_e c^2 \frac{j(j+1) - l(l+1) - \frac{3}{4}}{l(l+1)(2l+1)}$$

$$E_0 = \frac{\alpha^2}{2} m_e c^2$$

Rydberg (oen. energija) →

$$\langle E_{ls} \rangle = \frac{Z^4 \alpha^2}{\hbar^3} E_0 \frac{j(j+1) - l(l+1) - \frac{3}{4}}{l(l+1)(2l+1)}$$

Zgled

Stanja 2P ( $n=2, l=1, j = (\pm \frac{1}{2})$ )

$$j = \frac{3}{2}: \langle E_{ls} \rangle = \frac{\alpha^2}{8} E_0 \frac{\frac{3}{2} \cdot \frac{5}{2} - 1 \cdot 2 - \frac{3}{4}}{1 \cdot 2 \cdot 3} = \frac{\alpha^2 E_0}{48}$$

$$\rightarrow \begin{array}{c} n=2, l=1 \\ \text{-----} j = \frac{3}{2} \\ \text{-----} j = \frac{1}{2} \end{array}$$

$$j = \frac{1}{2}: \langle E_{ls} \rangle = \frac{\alpha^2}{8} E_0 \frac{\frac{1}{2} \cdot \frac{3}{2} - 1 \cdot 2 - \frac{3}{4}}{1 \cdot 2 \cdot 3} = -\frac{\alpha^2 E_0}{24}$$

celoten razcep:

$$\Delta E_{ls} = \frac{\alpha^2 E_0}{16} \approx 4,5 \cdot 10^{-5} \text{ eV}$$

## Relativistični popravek k kinetični energiji

$$T = \sqrt{p^2 c^2 + m_e^2 c^4} - m_e c^2 = m_e c^2 \sqrt{1 + \frac{p^2 c^2}{m_e^2 c^4}} - m_e c^2 = m_e c^2 \left( \cancel{1} + \frac{p^2}{2 m_e^2 c^2} - \frac{p^4}{8 m_e^4 c^4} + \dots \cancel{-1} \right) =$$

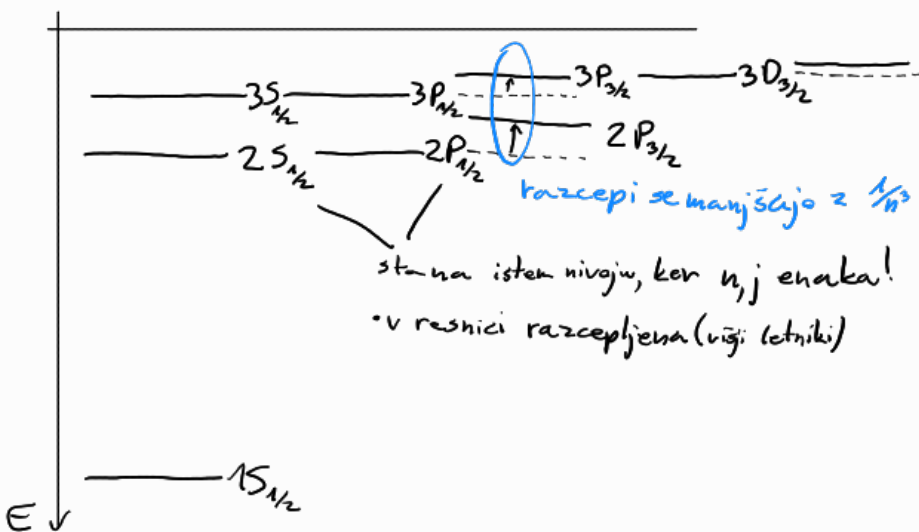
$$= \frac{p^2}{2 m_e} - \underbrace{\frac{p^4}{8 m_e^3 c^2}}_{\text{popravek, } T_{\text{rel}}} + \dots$$

$$\langle T_{\text{rel}} \rangle = - \frac{Z^4 \alpha^4}{n^3} m_e c^2 \left( \frac{1}{2l+1} - \frac{3}{8n} \right)$$

$$\rightarrow \langle E_{\text{ls}} \rangle + \langle T_{\text{rel}} \rangle = \frac{Z^4 \alpha^4}{2 n^3} m_e c^2 \left[ \underbrace{\frac{j(j+1) - l(l+1) - 3/4}{l(l+1)(2l+1)} - \frac{2}{2l+1} + \frac{3}{4n}}_{-\frac{2}{2j+1} \text{ neodv. od } l!} \right]$$

Energije stanj v H-atomu

$$E_{nj} = E_n - \frac{Z^4 \alpha^2}{n^3} \epsilon_0 \left[ \frac{2}{2j+1} - \frac{3}{4n} \right]$$



## ZEE MANOV POJAV

• atom v zeh. mag. polju

$$E_{\text{mag}} = -\vec{\mu} \cdot \vec{B}$$

• celoten mag. moment  $\vec{\mu} = \vec{\mu}_L + \vec{\mu}_S = -\frac{\mu_B}{\hbar} (g_L \vec{L} + g_S \vec{S}) = -\frac{\mu_B}{\hbar} (\vec{L} + 2\vec{S})$

$\vec{L} + \vec{S} = \vec{J}$

# 1. Močno polje ( $\vec{B} \gg \vec{B}_{int}$ )

• sklopitev LS lahko zanemarimo ( $\vec{L}$  in  $\vec{S}$  sta razklopljena, okrog  $\vec{B}$  precesirata vsak zase)

→  $m_l, m_s$  dobri KŠ

$$\langle L_z \rangle = m_l \hbar$$

$$\langle S_z \rangle = m_s \hbar$$

$$\vec{B} \parallel \hat{z}$$

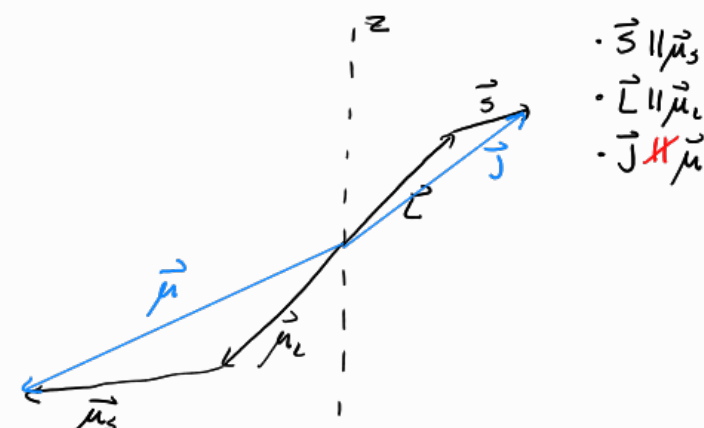


$$\langle E_{mag} \rangle = \frac{\mu_B}{\hbar} (\langle L_z \rangle + 2\langle S_z \rangle) B = \mu_B (m_l + 2m_s) B$$

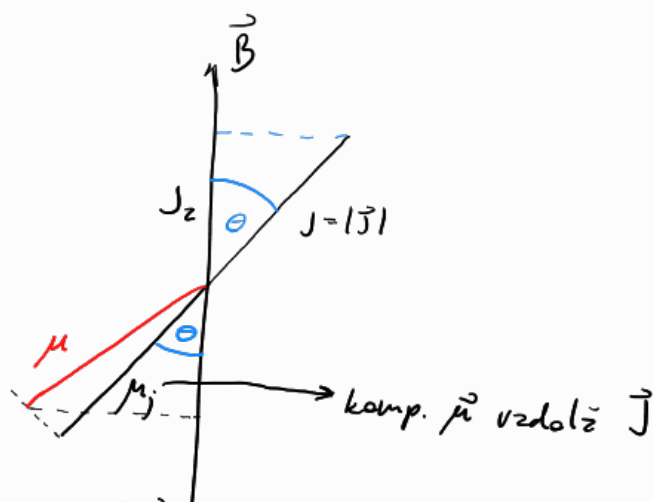
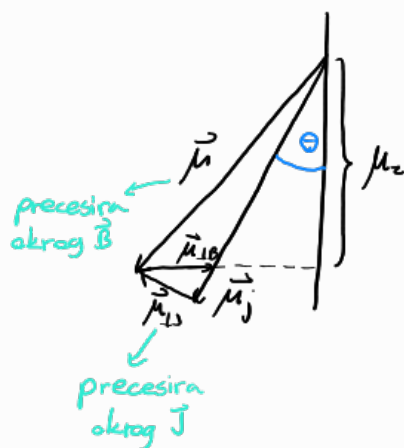
# 2. Šibko polje ( $\vec{B} \ll \vec{B}_{int}$ )

•  $\vec{L}$  in  $\vec{S}$  ostaneata sklopljena

→  $j, m_j$  dobri KŠ



- $\vec{S} \parallel \vec{\mu}_s$
- $\vec{L} \parallel \vec{\mu}_l$
- $\vec{J} \nparallel \vec{\mu}$



$$\mu_j = \frac{\vec{\mu} \cdot \vec{J}}{|\vec{J}|} = -\frac{\mu_B}{\hbar |\vec{J}|} (\vec{L} + 2\vec{S}) \cdot (\vec{L} + \vec{S}) =$$

$$= -\frac{\mu_B}{\hbar} (\vec{L}^2 + 2\vec{S}^2 + 3\vec{L} \cdot \vec{S}) =$$

$$\vec{L} \cdot \vec{S} = \frac{1}{2} (\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$$

$$= -\frac{\mu_B}{2\hbar} (3\vec{J}^2 + \vec{S}^2 - \vec{L}^2)$$

$$\mu_z = |\vec{\mu}_j + \vec{\mu}_{\perp j} + \vec{\mu}_{\perp S}| \cos \theta \rightarrow \cos \theta = \frac{J_z}{J}$$

po smereh se izpolnečeta  
v 0

$$\Rightarrow \mu_z = \mu_j \cos \theta = J_z \frac{\mu_j}{J}$$

Lahko bi računali  $\langle \mu_z \rangle$ , toda raje  $\langle \mu_z \rangle \vec{J}^2 = \langle J_z J \mu_z \rangle = -\frac{\mu_B}{2\hbar} \langle J_z (3\vec{J}^2 + \vec{S}^2 - \vec{L}^2) \rangle$

$$\langle \mu_z \rangle \hbar^2 j(j+1)$$

$$= \langle J_z \rangle (3\hbar^2 j(j+1) + \hbar^2 s(s+1) - \hbar^2 l(l+1))$$

$$\rightarrow \langle \mu_z \rangle = -g \mu_B \frac{\langle J_z \rangle}{\hbar}$$

$$g = \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

$$\rightarrow \langle \vec{\mu} \rangle = -g \mu_B \frac{\langle \vec{J} \rangle}{\hbar}$$

Landéjevo giramagnetno razmerje

$$\langle E_{\text{mag}} \rangle = -\langle \vec{\mu} \rangle \cdot \vec{B} = -\langle \mu_z \rangle B = g \mu_B \frac{\langle J_z \rangle}{\hbar} B = \underline{\underline{g \mu_B B}}$$

Zgled: prehodi  $2P_{3/2}, 2P_{1/2} \rightarrow 2S_{1/2}$

$$2P_{3/2}: g = 1 + \frac{\frac{3}{2} \cdot \frac{5}{2} - 1 \cdot 2 + \frac{3}{4}}{\frac{15}{2}} = \frac{4}{3}$$

$$\langle E_{\text{mag}} \rangle = \frac{4}{3} \mu_B B \cdot \left\{ \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2} \right\}$$

Pri prehodu se m.j. lahko spremeni za največ 1

$$2P_{1/2}: g = 1 + \frac{\frac{1}{2} \cdot \frac{3}{2} - 1 \cdot 2 + \frac{3}{4}}{\frac{3}{2}} = \frac{2}{3}$$

$$\langle E_{\text{mag}} \rangle = \frac{2}{3} \mu_B B \cdot \left\{ \frac{1}{2}, -\frac{1}{2} \right\}$$

$$1S_{1/2}: g = 1 + \frac{\frac{1}{2} \cdot \frac{3}{2} - 0 + \frac{3}{4}}{\frac{3}{2}} = 2$$

$$\langle E_{\text{mag}} \rangle = 2 \mu_B B \cdot \left\{ \frac{1}{2}, -\frac{1}{2} \right\}$$

## Sevanje atomov

• doslej smo lastna stanja (rešitve SSE za H-atom) obravnavali kot stacionarna:

$$|\Psi_n(\vec{r}, t)|^2 = |\Psi_n(\vec{r})|^2$$

• zaradi motenj atom ni več v stac. st.

↳ trki, zunanje EM motnje, končna razsežnost jedra, ...

→ obravnavamo prehode iz 1. vzb. st. (2) v osk. st. (1)

(izpeljava ok za H-atom, α-jama, LHO, ...)

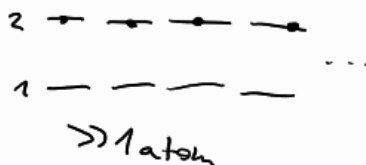
$$\Psi_1(\vec{r}, t) = \Psi_1(\vec{r}) e^{-iE_1 t/\hbar}, \quad \Psi_2(\vec{r}, t) = \Psi_2(\vec{r}) e^{-iE_2 t/\hbar}$$

• če želimo, da pride do prehoda, mora ∃ še neka druga čas. odv. poleg eksponenta

• mislimo si tako VF:

$$\Psi_\alpha = \begin{cases} \Psi_2; & t \leq 0 \\ \Psi_1; & t > \tau \end{cases}$$

↑  
karakteristični  
čas za razpad  
vzb. stanja /  
atoma v vzb. st.



$$\Psi_\alpha = c_1(t) \Psi_1 + c_2(t) \Psi_2$$

$c_1, c_2 \in \mathbb{C}$ , toda recimo  $\mathbb{R}$

velja  $c_1^2 + c_2^2 = 1$

$$\begin{aligned} \langle E_\alpha \rangle &= c_1^2(t) E_1 + c_2^2(t) E_2 = (1 - c_2^2(t)) E_1 + c_2^2(t) E_2 = \\ &= E_1 + c_2^2(t) [E_2 - E_1] \end{aligned}$$

-pred razpadom:  $c_1=0, c_2=1$

-po razpadu:  $c_1=1, c_2=0$

→ obravnavamo el. dipolno sevanje

• jedro je  $\approx$  fiksno  $\rightarrow \hat{p}_e = -e_0 \vec{r}$

• pričakovana vrednost  $\vec{p}_e$  v stanjih  $\Psi_\alpha$ :

$$\begin{aligned} \langle \vec{p}_e(\alpha) \rangle &= \int \Psi_\alpha^* \hat{p}_e \Psi_\alpha dV = \int (c_1 \Psi_1^* + c_2 \Psi_2^*) (-e_0 \vec{r}) (c_1 \Psi_1 + c_2 \Psi_2) d^3\vec{r} = \\ &= c_1^2 \int \Psi_1^* \hat{p}_e \Psi_1 d^3\vec{r} + c_2^2 \int \Psi_2^* \hat{p}_e \Psi_2 d^3\vec{r} + c_1 c_2 \int \Psi_1^* \hat{p}_e \Psi_2 d^3\vec{r} + c_1 c_2 \int \Psi_2^* \hat{p}_e \Psi_1 d^3\vec{r} \end{aligned}$$

\* vsen integralom tipa  $\int \Psi_1^* \hat{p}_e \Psi_2 dV$  pravimo matrični elementi

→ diagonalni matrični elementi  $\rightarrow 0$ , ker nastopajo elementi oblike  $\int |\Psi_\alpha|^2 x dV$   
in imamo pod integralom produkt sode in lihe funkcije  
ali  $y, z$

$$\vec{p}_e^{(12)}(t) = \int \Psi_1^* \hat{p}_e \Psi_2 d^3\vec{r} = e^{-i(\epsilon_2 - \epsilon_1)t/\hbar} \underbrace{\int \Psi_1^*(\vec{r}) \hat{p}_e \Psi_2(\vec{r}) d^3\vec{r}}_{\vec{p}_e^{(12)} \rightarrow \text{brez}(t)}$$

→ predp., da  $\vec{p}_e^{(12)} = \vec{p}_e^{(21)}$  (da dobimo  $\cos$ , in ne  $\cos(-15^\circ)$ )

$$\rightarrow \langle \vec{p}_e(\alpha) \rangle = c_1 c_2 \vec{p}_e^{(12)} \left[ e^{-i(\epsilon_2 - \epsilon_1)t/\hbar} + e^{i(\epsilon_2 - \epsilon_1)t/\hbar} \right] = 2 c_1 c_2 \vec{p}_e^{(12)} \cos\left(\frac{(\epsilon_2 - \epsilon_1)t}{\hbar}\right)$$

↑  
ključni S  $\omega_{12}$

$\Rightarrow \langle \vec{p}_e(\alpha) \rangle$  harmonsko nihanje s krožno frekvenco  $\omega_{12} = \frac{(\epsilon_2 - \epsilon_1)t}{\hbar}$

energ. tok  
↓  
 $\rho = \frac{\omega^2 \vec{p}_{e0}^2}{12\pi\epsilon_0 c^3}$

$\vec{p}_e(t) = \vec{p}_{e0} \cos(\omega t)$

$\rho = -\frac{d\epsilon}{dt}$

→ ko je večina atomov že v osh. st.:

$$c_1 < 1, c_2 = \sqrt{1 - c_1^2} \ll 1$$

$$\rho = -\frac{d}{dt} \langle E_\alpha \rangle = -\frac{d}{dt} c_2^2 (\epsilon_2 - \epsilon_1) = -\hbar \omega_{12} \frac{dc_2^2}{dt}$$

$$\rho = \frac{\omega^2 \vec{p}_{e0}^2}{12\pi\epsilon_0 c^3} \rightarrow \rho = \frac{\omega_{12}^2 (2c_1 c_2 \vec{p}_e^{(12)})^2}{12\pi\epsilon_0 c^3} = -\hbar \omega_{12} \frac{dc_2^2}{dt}$$

$$\rightarrow \frac{dc_2^2}{c_2^2} = - \left( \frac{\omega_{12}^2 (\vec{p}_e^{(12)})^2}{3\pi\epsilon_0 c^3 \hbar} \right) dt \rightarrow c_2^2(t) = \underbrace{c_2^2(0)}_{=1} e^{-t/\tau}$$

$\tau = 1/\gamma$

$$C_2^2(t) = e^{-\frac{t}{\tau}}$$

$$\frac{1}{\tau} = \frac{\omega_{21}^2 [\frac{\vec{p}_e^{(21)}}{p_0}]^2}{3\pi \epsilon_0 c^3 \hbar}$$

št. stanj, ki so še na voljo  
za razpad, eksp. pada

## Izbirna pravila

→ povedo, kdaj je razpad/prehod dovoljen

Če vsaj 1 komponenta  $\int \Psi_1^* (-e \vec{r}) \Psi_2 d^3 \vec{r}$  nenulna  $\Rightarrow$  prehod dovoljen

Sicer je prehod prepovedan.

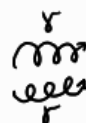
$$\Delta m_l = 0, 1, -1$$

$$\Delta l = \pm 1$$

↳ foton, ki odnese  $E$ ,  
ima spin 1

→ projekciji  $m_s = \pm 1$  (desno/levo krožno polarizirano)

• realni fotoni:  $\vec{E}_r^2 - \vec{p}_r^2 c^2 = 0$



$$\Delta m_s = 0$$

$$\Delta s = 0$$

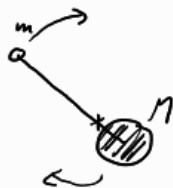
Ler operator  $\vec{r}$   
ne more spremeniti  
spina

$m_s = 0$  (longitudinalno polarizirano)

• virtualni fotoni:  $\vec{E}_r^2 - \vec{p}_r^2 c^2 \neq 0$

- realni fotoni so samo tisti, ki letijo  $\infty$  časa (ampak je že npr. relacija Sonce-Zemlja dober približek)

## Rotator



$$W_{\text{rot}} = \frac{\vec{L}^2}{2J} = \frac{\hbar^2 l(l+1)}{2\mu r^2}$$

$$\Delta l = \pm 1$$

ker drugih prost. stopenj ni

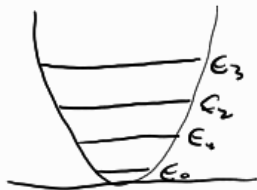
## LHO

$$E_n = \hbar \omega_0 (n + \frac{1}{2})$$

$\infty$  jama

$$\Delta n = \text{liha}$$

$$\Delta h = \pm 1$$



## Širina spektralnih črt

Heisenberg:  $\Delta E \Delta t \geq \frac{\hbar}{2}$

Dagovar:  $E_{1/2} \tau = \hbar$   
 karakt. čas prehoda  
 razpolovna širina stanja

## Klasično:

El. polje: frekv. oscilacije  
 ↓  
 dušenje

$$E(t) = E_0 e^{-i\omega_0 t - \beta t}$$

Fourierova transf.  $E(t) \rightarrow \tilde{E}(\omega)$

$$\tilde{E}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} E(t) e^{i\omega t} dt = \frac{E_0}{\sqrt{2\pi} (\beta + i(\omega_0 - \omega))}$$

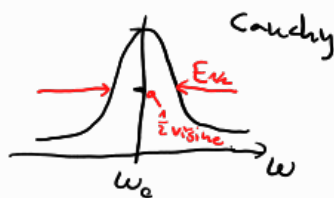
$$\text{spekter} = |\tilde{E}(\omega)|^2 \propto \frac{1}{(\omega - \omega_0)^2 + \beta^2}$$

## Kvantno:

$$\beta \rightarrow 1/\tau$$

$\hbar\omega_0 \rightarrow E_2 - E_1$  (centralna vrednost energije prehoda)

$\hbar\omega \rightarrow E_\gamma$  (energija, ki jo foton odnese)



$$P(\epsilon) = \frac{1}{\pi} \frac{\frac{\hbar}{2\tau}}{(\epsilon - E_0)^2 + (\frac{\hbar}{2\tau})^2}$$

$\tau$  velik  $\rightarrow$  črta ostrejša ( $E_{1/2}$  majhna)  
 (stanje dolgoživo)

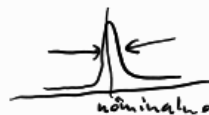
$E_{1/2} \rightarrow$  naravna širina spektr. črt

Ocena: H-atom,  $n=2 \rightarrow n=1$

$$E_2 - E_1 = \frac{3}{4} 13,6 \text{ eV}$$

$$1/\tau = 6,3 \cdot 10^8 \text{ s}^{-1} = \omega_{1/2}$$

$$E_{1/2} = \frac{\hbar}{2} 6,3 \cdot 10^8 \text{ eV}$$



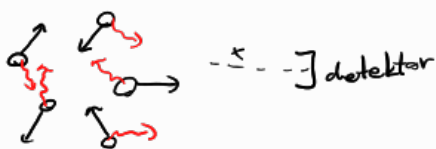
Rel. širine so v vseh merilih enake:

$$\frac{E_{1/2}}{E} = \frac{\omega_{1/2}}{\omega} = \frac{v_{1/2}}{v} = \frac{\lambda_{1/2}}{\lambda} = 4 \cdot 10^{-8}$$

$$v_{1/2} = \frac{1}{2\pi\tau} = 1,0 \cdot 10^8 \text{ s}^{-1}, \quad \lambda_{1/2} = \frac{1}{c} v_{1/2} = 4,9 \cdot 10^{-6} \text{ nm}$$

V resnici vidimo bistveno širše črte zaradi Dopplerja in trkov.

## 1) Doppler



$$\sigma\omega_D = \frac{v_x}{c} \omega \rightarrow \langle v_x \rangle = 0, \text{ bolje vzeti } \sqrt{\langle v^2 \rangle} = \sqrt{\frac{k_B T}{m}}$$

$$\sigma\omega_D \approx \frac{1}{c} \sqrt{\frac{k_B T}{m}} \omega_0$$

Dopplerjeva razširitev spektralnih črt

Ocena: vodik pri sobni T

$$\sigma\omega_D \approx 1,5 \cdot 10^{11} \text{ s}^{-1}$$

$$\sigma\lambda_D \approx 2 \cdot 10^{-3} \text{ nm} \quad (\sigma\lambda_{\text{naravna}} \approx 10^{-6} \text{ nm})$$

## 2) Trki

$\delta \omega_T \approx \frac{1}{\tau_T}$  atom, ki seva, lahko to stori le v časih med dvema trkoma

ocena:  $\tau_T \approx \frac{\langle l \rangle}{\langle v \rangle}$

$$\langle l \rangle \approx \frac{1}{\pi (2r)^2 n}$$

$$\uparrow$$

$$\rho = nk_B T$$

$$\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}}$$

$$\delta \omega_T \approx 2 \sqrt{\frac{2\pi}{mk_B T}} (2r)^2 \rho$$

$\uparrow$   
tlak

razširitev zaradi  
trkov

Ocena: sobna T,  $p = 1 \text{ bar}$ ,  $r = 0,14 \text{ nm}$

$$\delta \omega_T = 1,4 \cdot 10^{10} \text{ s}^{-1}$$

$$\delta \lambda_T = 1,1 \cdot 10^{-4} \text{ nm}$$